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A transfer-matrix approach to one-dimensional quantum mechanics using Mathematica

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A variety of one-dimensional quantum systems is studied using a transfer-matrix formalism. In particular, only two matrices are required: one to propagate a wave function over a region of constant potential, the other to connect wave functions at a discontinuity in the potential. Using these simple matrices as building blocks, we constructed complex transfer matrices with which to study energy bands in one-dimensional crystals, as well as the intriguing characteristics of transmission through multiple barriers. All systems studied here are modeled with finite-step potentials, not delta functions. The solutions obtained are exact and can be generated with relative ease using Mathematica. The method is thoroughly explicated and several coding examples are provided.

I. INTRODUCTION

Transfer matrices provide a powerful and compact formalism for the study of physical systems occurring in many areas of physics, including statistical mechanics,¹ optics,² and quantum mechanics.³ In one-dimensional quantum mechanics, for example, transfer matrices produce exact solutions for systems with potentials in the form of an arbitrary series of steps and plateaus. Since any one-dimensional potential can be approximated to arbitrary precision with steps and plateaus of appropriate size, the transfer-matrix method is suitable for all one-dimensional quantum systems.

The transfer-matrix formalism developed here is different from those presented elsewhere,^{3,4} in part through the introduction of two simple matrices—the propagation matrix and the discontinuity matrix—which are used to construct the transfer matrix for any given system. As a result, the method is quite straightforward to apply. In addition, the ease with which Mathematica⁵ produces graphics and handles matrix manipulations, even when the matrix elements are complex, makes the actual implementation of the method a pleasure.

In this paper we restrict our attention to rectangular potentials, such as those shown in Fig. 1, and thus all of our results are exact. To illustrate the method, we first consider the energy bands of a one-dimensional crystal formed of an infinite array of barriers. We then investigate the transmission characteristics of a finite series of barriers. Though results for multiple barriers are practically nonexistent,⁶ due to the difficulty of solving such a system with traditional techniques, our method makes the study of any number of barriers as simple as the study of a single barrier. This allows us to present new results for multiple barriers, results that elucidate the physical origin of band structure in crystals.

II. TRANSFER-MATRIX FORMALISM

In this section we develop a transfer-matrix approach to one-dimensional quantum mechanics. In particular, we derive two simple matrices—one for propagating a wave function over regions of constant potential, and one for connecting wave functions at discontinuities in the potential. These basic matrices serve as fundamental units to be used in constructing the transfer matrix for any given one-dimensional system.

We consider first the propagation matrix, associated with regions of constant potential. In particular, suppose the potential $V(x)$ in Schrödinger's equation

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi(x)}{dx^2} = [E - V(x)] \psi(x) \quad (1)$$

is a positive constant V , in which case the wave function is given by

$$\psi(x) = A e^{ikx} + B e^{-ikx}, \quad (2)$$

where $k = \sqrt{E/V - 1} \sqrt{2mV/\hbar}$. A convenient way to write the wave function—for the transfer-matrix formalism to follow—is as a vector dot product:

$$\psi(x) = (e^{ikx} \ e^{-ikx}) \begin{pmatrix} A \\ B \end{pmatrix} = (e^{ikx} \ e^{-ikx}) \varphi, \quad (3)$$

with the coefficient vector φ defined by $\varphi = \begin{pmatrix} A \\ B \end{pmatrix}$.

Now, consider a translation of the coordinate system to the right a distance L , producing a translated system with coordinate x' given by $x' = x - L$. It is assumed, of course, that the point $x = L$ also lies within the region of constant potential V . The condition that both systems describe the same wave function is $\psi'(x') = \psi(x)$. Noting that $x = x' + L$, we find

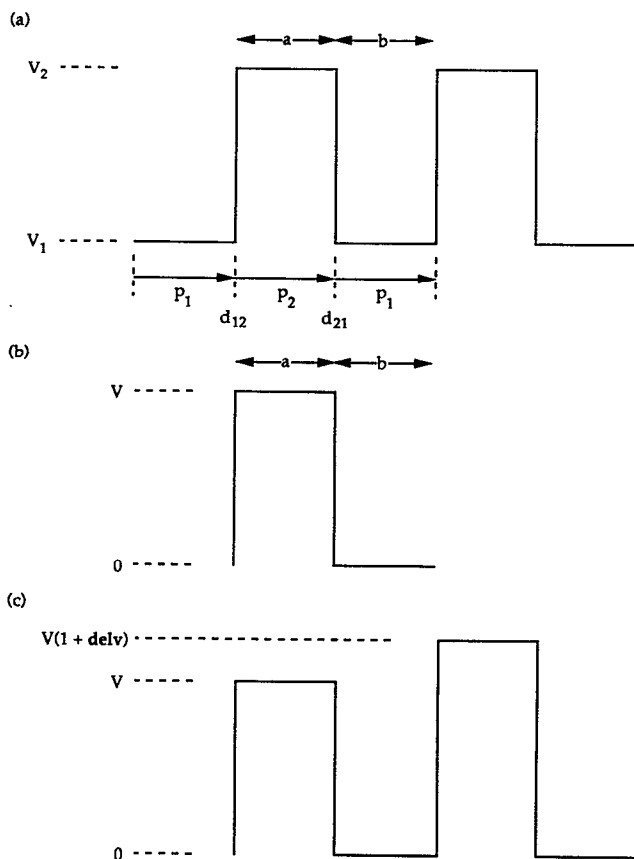


FIG. 1. (a) Potential energy function illustrating notation used in the transfer-matrix formalism. (b) Unit cell for a simple one-dimensional crystal in which all barriers have the same height V . The length of the unit cell is $d = a + b$. (c) Unit cell for a one-dimensional crystal in which barrier heights alternate between the values V and $V(1 + \text{delv})$. In this case the unit-cell length is $2d$.

$$\begin{aligned}\psi'(x') &= \psi(x' + L) = (e^{ikx'} e^{-ikx'}) \begin{pmatrix} Ae^{ikL} \\ Be^{-ikL} \end{pmatrix} \\ &= (e^{ikx'} e^{-ikx'}) \varphi',\end{aligned}\quad (4)$$

where $\varphi' = \begin{pmatrix} A' \\ B' \end{pmatrix}$ with $A' = Ae^{ikL}$ and $B' = Be^{-ikL}$. This transformation can be expressed most simply in terms of a matrix p , the propagation matrix, defined by $\varphi' = p\varphi$. Clearly,

$$p = \begin{pmatrix} e^{ikL} & 0 \\ 0 & e^{-ikL} \end{pmatrix} \quad (5)$$

is the explicit expression for p .

In the program code given in the Appendices the propagation matrix for propagating a distance D over a region where the potential is V_i is defined as $p[\text{vi}, \text{length}]$, where $\text{vi} = V_i/V$ is the dimensionless potential, and $\text{length} = D/a$ is the dimensionless distance of propagation. In what follows, we take the distance a to be the width of a barrier, as indicated in Fig. 1(a). Also, we normalize all energies with the positive potential V , thus in regions where the potential is V_i the associated wave number k_i is written as $k_i = \sqrt{E/V - V_i/V}$

$\times \sqrt{2mV}/\hbar$. Note that the definitions for the propagation matrix and k_i are equally valid for both $E > V_i$ and $E < V_i$; in the latter case k_i is complex and the exponentials in p are real, as expected. Finally, if a given propagation matrix is $p = p[\text{vi}, \text{length}]$, the corresponding inverse propagation matrix is simply $p^{-1} = p[\text{vi}, -\text{length}]$.

With appropriate use of the propagation matrix it can always be arranged that a discontinuity in the potential occurs at the origin. Assume, then, a discontinuity at $x = 0$, with the potential defined by $V(x) = V_1$ for $x < 0$ and $V(x) = V_2$ for $x > 0$. Since the potential is constant on either side of the origin, it follows that the wave function is

$$\begin{aligned}\psi_1(x) &= A_1 e^{ik_1 x} + B_1 e^{-ik_1 x} \quad (x < 0), \\ \psi_2(x) &= A_2 e^{ik_2 x} + B_2 e^{-ik_2 x} \quad (x > 0),\end{aligned}\quad (6)$$

where $k_1 = \sqrt{E/V - V_1/V} \sqrt{2mV}/\hbar$ and $k_2 = \sqrt{E/V - V_2/V} \sqrt{2mV}/\hbar$. To be a valid solution of Schrödinger's equation, the wave function and its first derivative with respect to x must be continuous everywhere. Applying these boundary conditions at $x = 0$ yields

$$\begin{aligned}A_1 + B_1 &= A_2 + B_2, \\ ik_1(A_1 - B_1) &= ik_2(A_2 - B_2).\end{aligned}\quad (7)$$

Defining $\varphi_1 = \begin{pmatrix} A_1 \\ B_1 \end{pmatrix}$ and $\varphi_2 = \begin{pmatrix} A_2 \\ B_2 \end{pmatrix}$, we can express the boundary conditions in matrix form as $\varphi_2 = d_{12}\varphi_1$, where

$$d_{12} = \frac{1}{2} \begin{pmatrix} 1 + k_1/k_2 & 1 - k_1/k_2 \\ 1 - k_1/k_2 & 1 + k_1/k_2 \end{pmatrix} \quad (8)$$

is the discontinuity matrix connecting regions of potential V_1 and V_2 .

In the program code the discontinuity matrix is calculated with $d[\text{vi}, \text{vj}]$, where $\text{vi} = V_i/V$ and $\text{vj} = V_j/V$. For example, the matrix defined above is simply $d_{12} = d[\text{v1}, \text{v2}]$, and its inverse is $(d_{12})^{-1} = d[\text{v2}, \text{v1}] = d_{21}$.

We are now in a position to construct transfer matrices using the two matrices p and d . As an example, consider the potential shown in Fig. 1(a). Clearly, the propagation matrices for regions where the potential is V_1 and V_2 are $p_1 = p[\text{v1}, b/a]$ and $p_2 = p[\text{v2}, 1]$, respectively. Similarly, the discontinuity matrices for going from V_1 to V_2 and from V_2 to V_1 are $d_{12} = d[\text{v1}, \text{v2}]$ and $d_{21} = d[\text{v2}, \text{v1}]$, respectively. Thus, if φ_{left} represents the coefficient matrix at the left side of Fig. 1(a), the coefficient matrix at the right side of the figure, φ_{right} , is given by

$$\begin{aligned}\varphi_{\text{right}} &= [p_1 d_{21} p_2 d_{12} p_1 d_{21} p_2 d_{12} p_1] \varphi_{\text{left}} \\ &= t \varphi_{\text{left}},\end{aligned}\quad (9)$$

which defines the transfer matrix t for this potential.

In the sections that follow, we consider several examples of one-dimensional potentials, and in each case obtain the corresponding transfer matrix. We show that physical significance can be associated with several aspects of a transfer matrix, including individual matrix elements and the magnitude and phase of the eigenvalues. Although not discussed in this paper, it can be shown that the eigenvectors also contain a wealth of physical information.

III. ONE-DIMENSIONAL CRYSTAL

We now apply the method outlined above to a standard textbook problem:^{7,8} the behavior of an electron in a one-

dimensional crystal lattice. The periodic potential representing the lattice is modeled as an infinite succession of rectangular barriers, that is, simply an extension in either direction of the potential shown in Fig. 1(a). In particular, we consider first the special case⁹ in which

$$V_1 = 0, \quad V_2 = V, \quad a = b,$$

and

$$\rho = a\sqrt{2mV}/\hbar = \pi/0.9.$$

Figure 2(a) presents the allowed values of E/V for such a lattice, a result familiar from discussions in many quantum mechanics texts. Note especially that for the horizontal axis we use kd/π , where k is the wave number and $d = a + b$.

One would expect a transfer-matrix implementation to employ some process involving successive matrix multiplications, in this case propagating the electron through

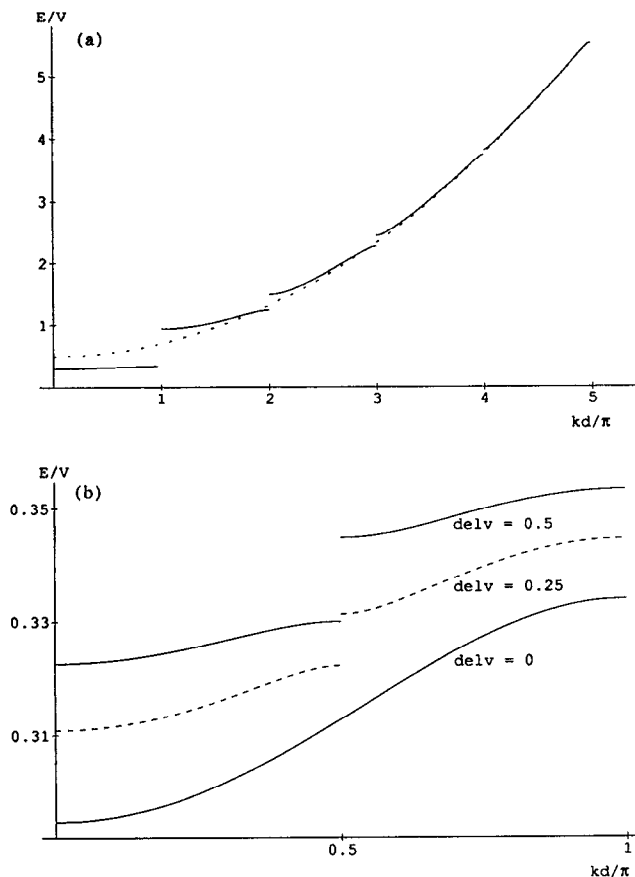


FIG. 2. (a) Allowed energies in a one-dimensional periodic lattice of rectangular potentials. The horizontal axis is kd/π , where k is the wave number and d the length of the repeating pattern of barriers and gaps. The pattern used for this figure is pictured in Fig. 1(b). The dotted line is the curve for the free electron, displaced upward by $1/2$. (b) Band splitting. If the barriers are made to alternate in height, with the potential difference between consecutive barriers given by ΔV , the energy bands will appear to split. The thick curve on the bottom, $\Delta V = 0$, is simply a blowup of the first band in (a); $\Delta V = 0.25$ produces the dashed bands; $\Delta V = 0.5$ results in the bands on the top. Figure 1(c) depicts the alternating pattern of potentials that generated the split bands. Note that the average value of the potential increases with ΔV , hence the increasing trend in the band energies.

barrier after barrier after barrier. For example, suppose we are given the transfer matrix t_n that propagates an input vector v_{in} across the n th barrier. To get the electron across four barriers would then require the following operation:

$$v_{out} = t_4 t_3 t_2 t_1 v_{in}. \quad (10)$$

In general, to propagate the electron across n barriers requires n transfer matrices, which is indeed the technique adopted below in the discussion concerning the transmission coefficient of a system composed of some relatively small number of barriers. One might be tempted to extend this line of attack to the crystal lattice by carrying out a large number of matrix multiplications, to approximate with many barriers the properties of a (nearly) infinite array; but the method presented here allows a more interesting and elegant solution.

We first note that if all the transfer matrices are identical—which is certainly true here, given our definition of a unit cell—then, with

$$t_{uc} = p_1 d_{21} p_2 d_{12} \quad (11)$$

defining the transfer matrix for the unit cell depicted in Fig. 1(b), Eq. (10) can be generalized for n unit cells as

$$v_{out} = t_{uc}^n v_{in}. \quad (12)$$

Furthermore, if λ is an eigenvalue of t_{uc} , then λ^n is an eigenvalue of t_{uc}^n . Since the eigenvalue expresses the factor by which the propagating wave vector changes as it traverses the region represented by the constructed unit cell matrix, any E/V for which the magnitude of the eigenvalue is other than unity will represent a disallowed energy, as it will either be beaten down to zero or blow up to infinity as $n \rightarrow \infty$. So, to find the allowed energies, we simply use t_{uc} : An electron with an E/V yielding a t_{uc} whose eigenvalues are of magnitude 1 will propagate through the lattice; others will not.

Now that we have used the magnitude of the eigenvalue to delineate the energy bands, we can also use it to determine kd by looking at its argument. The argument of the eigenvalue provides the phase factor by which the wave has changed in propagating across the unit cell, and this, of course, is exactly what is meant by kd . (Actually, the phase factor is positive or negative, depending on which eigenvalue is considered: One is associated with motion to the right, the other with motion to the left.) Thus both axes in Fig. 2(a) can be obtained with a surprisingly simple and quick algorithm, analyzing only information about the magnitude and argument of the eigenvalues.

The dashed line component of Fig. 2(a) represents an interesting feature of the graph. It is the limiting curve of the E/V vs kd/π plot; that is, the energy bands asymptotically approach this curve as $E/V \rightarrow \infty$. In essence, it is the free electron model result, displaced upward by $1/2$.

To briefly explore the nature of this result, we shall start with the dispersion relation,⁷ written in terms of the dimensionless quantity ρ defined earlier. Letting

$$e = E/V,$$

$$A = \rho e^{1/2},$$

$$B = \rho(e - 1)^{1/2},$$

$$C = (2e - 1)/[2(e^2 - e)]^{1/2},$$

the dispersion relation may then be expressed as

$$\cos(kd) = \cos(A)\cos(B) - C\sin(A)\sin(B). \quad (13)$$

In the limit $e \rightarrow \infty$, we see that $B \rightarrow A$ and $C \rightarrow 1$. This lets us rewrite Eq. (13), obtaining

$$\cos(kd) = \cos(2A). \quad (14)$$

From Eq. (14) we find the asymptotic form of the free electron energy curve,

$$E/V = (1/4\rho^2)(kd)^2. \quad (15)$$

Whence the upward displacement of $1/2$, which does not appear in Eq. (15)? This offset is rigorously accounted for in a more complete expansion of the dispersion relation; physically one might think of the electron as “riding high” over the potential landscape and thus experiencing only the average barrier height, $1/2$ in this case. The reader is invited to experiment with this by changing either the barrier height or length of the gap between barriers ($\sqrt{2}$ and length, respectively, in Appendix A).

The reader may also wish to compare Fig. 2(a) with Fig. 13-10 in Eisberg and Resnick,⁸ and with Fig. 8.13(a) in Liboff,⁷ which exhibit misleading characteristics for both the energy bands themselves as well as the relationship between the bands and the limiting curve.

Figure 2(b) illustrates the phenomenon known as “band splitting;” the split in the energy bands comes about in this case by arranging two, alternating barrier heights, as shown in Fig. 1(c). The `delv` term appearing in Fig. 2(b) determines the difference in the heights. With $p_2' = p[\sqrt{2} + \text{delv}, 1]$, $d_{12}' = d[\sqrt{1}, \sqrt{2} + \text{delv}]$, and $d_{21}' = d[\sqrt{2} + \text{delv}, \sqrt{1}]$, the transfer matrix in this case is

$$t_{uc}' = p_1 d_{21}' p_2' d_{12}' p_1 d_{21} p_2 d_{12}$$

(see Appendix B. To produce the band splitting shown in Fig. 2(b), set `period` = 2; `delv` = 0.25 or 0.5; and execute `searchBand`, as discussed below.)

The unit cell constructed in this process is now twice as long as the previous one, as can be seen by comparing Fig. 1(b) with 1(c). This doubling of the unit cell length means that the splitting occurs at 0.5 along the horizontal axis of Fig. 2(b), since the band edges occur whenever the wave number times the unit cell length is equal to a multiple of π .

Finally, Appendix B is a listing of the Mathematica code that constructs the matrix representing a unit cell. The routine `searchBand` in Appendix C generates a list of coordinates in the form $\{ \{kd/\pi, E/V\}, \{kd/\pi, E/V\}, \dots \}$, which can then be plotted with Mathematica's `ListPlot` facility to produce the energy bands shown in Fig. 2. The `searchBand` routine is fairly straightforward. It sweeps through a region of E/V values bounded by `start` and `stop`, with a resolution calculated from `step`.

IV. MULTIPLE BARRIERS

The previous section demonstrates the power of the transfer-matrix method, showing that the transfer matrix for a single unit cell, t_{uc} , determines the allowed energies for an

infinite crystal. Another strength of the method is that it simplifies the study of a finite, though possibly large, number of barriers to an analysis of the transfer matrix for a single unit cell raised to a power equal to the number of barriers. In what follows, we calculate the transmission coefficient for a wave of energy E incident on an array of barriers of height V . The “strength” of the barriers is given by $\rho = \pi/0.9$; thus, as the number of barriers becomes large, we see an approach to the infinite-lattice results of Sec. III.

To begin, consider a single barrier, for which the transfer matrix is $t = t_{uc}$. Imagine a wave incident from the right, which, upon encountering the barrier, produces both a reflected and a transmitted wave. In this case, the coefficient vector to the right of the barrier is $\varphi_{\text{right}} = \begin{pmatrix} A \\ B \end{pmatrix}$, where A and B are the amplitudes of the reflected and incident waves, respectively. Clearly, the transmitted wave consists solely of a wave traveling to the left; hence the coefficient vector to the left of the barrier can be written as $\varphi_{\text{left}} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. It follows that

$$\varphi_{\text{right}} = t \varphi_{\text{left}} = \begin{pmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} t_{12} \\ t_{22} \end{pmatrix}. \quad (16)$$

The transmission coefficient T is the mod squared ratio of the transmitted to the incident amplitude; thus $T = |1/t_{22}|^2$ for this system, utilizing just one element of the transfer matrix.

Figure 3(a) shows the results for T vs E/V for a single barrier, a standard result given in most quantum mechanics texts. Note that $T < 1$ for all $E/V < 1$, and that T approaches unity as $E/V \rightarrow \infty$. In addition, there are special energies where $T = 1$ exactly, corresponding to wave functions for which an integral number of half-wavelengths fit within the barrier region. These points of unit transmission are known as Ramsauer peaks,⁷ and, clearly, if a Ramsauer peak occurs at a given energy for a single barrier, it occurs at the same energy for multiple barriers as well.

Next, we consider the transmission coefficient for two barriers, $t = t_{uc}^2$, with the results shown in Fig. 3(b). It is clear immediately that qualitatively new behavior appears in this system; in fact, with only two barriers, we see already the first indications of the band structure observed in the one-dimensional crystal. Most striking is the extremely narrow peak indicating unit transmission at $E/V = 0.313$; that is, the barriers appear completely transparent to a beam with an energy only about a third of the barrier height. A second peak of unit transmission occurs at an energy just slightly above the barriers, at $E/V = 1.04$. The first Ramsauer peak occurs well above the barrier, at $E/V = 1.81$. These three peaks, in the order given, lie within the first, second, and third energy bands of the crystal, respectively.

To understand the origin of the first two unit transmission peaks, note that the region bounded by the two barriers is very much like a finite potential well, with a “strength” given by ρ . Such a well has two bound states, with energies of roughly 0.3 and 0.9 , in close correspondence with the first two peak energies. Of course, in the infinite crystal an infinite number of such wells interact to produce continuous energy bands associated with each of the bound states.

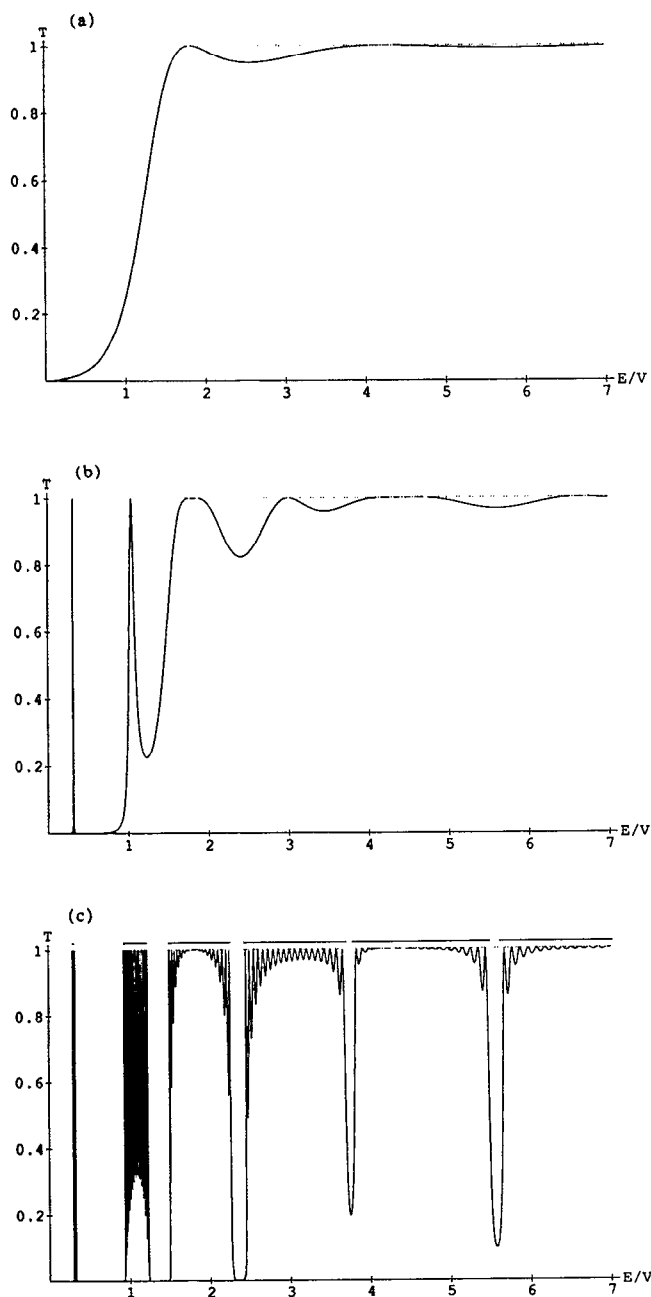


FIG. 3. Transmission coefficient T vs E/V for systems consisting of n identical barriers of height V . (a) $n = 1$; (b) $n = 2$. In this case new features appear, particularly the narrow peaks giving unit transmission for $E/V = 0.313$ and 1.04 . With just two barriers the band structure of the infinite crystal is already evident. (c) $n = 20$. In this system 19 peaks crowd into the region near $E/V = 0.3$, and 19 more occur near $E/V = 1$. The peaks of unit transmission are beginning to fill in continuous bands of energy, leaving gaps of near zero transmission. Line segments above $T = 1$ indicate energy bands of the infinite crystal.

Finally, we consider a system with 20 barriers, for which $t = t_{uc}^{20}$. The results, shown in Fig. 3(c), are a further extension of the behavior displayed by two barriers, showing regions of peaks reaching to unit transmission separated by regions of near zero transmission. The solid line

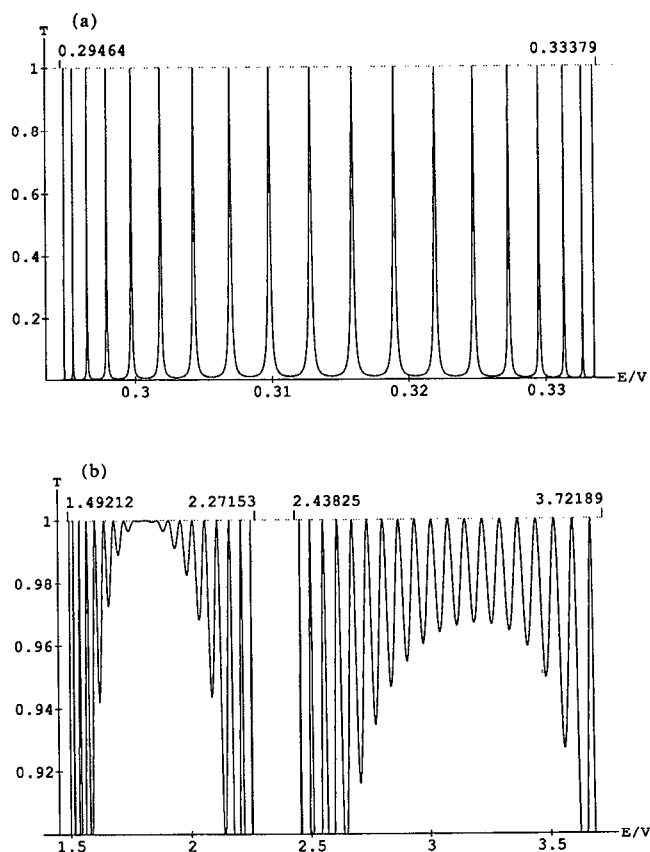


FIG. 4. (a) A detailed look at the dependence of transmission coefficient T on energy E/V , for a 20-barrier configuration. Note that even though the energies involved are all below the barrier height, there are still 19 energies that result in unit transmission. The energy values indicated along the top mark the edges of band 1. (b) The same information as in (a), for the range of energies corresponding to bands 3 and 4.

segments above the $T = 1$ line show the allowed energies corresponding to the first six energy bands of the crystal. Note that the peaks are filling in the full extent of each band, showing nicely the approach of a system with many barriers to the behavior of an infinite one-dimensional crystal.

Figure 4 provides a more detailed look at the results for 20 barriers. The transmission coefficient T is graphed in Fig. 4(a) for a range of E/V values corresponding to the first energy band. Figure 4(b) shows the same plot for bands three and four. Since 20 barriers create 19 wells, the first 2 bands have 19 peaks each. [Note from Fig. 3(c) that band two is comprised of peaks both inside and outside the wells.] Though not visible at the scale of the figure, band three actually has 20 peaks, with the extra peak due to the Ramsauer effect. Clearly, the presence of the Ramsauer peak significantly alters the transmission characteristics in that region. All odd bands from band three upward have 20 peaks; all even bands have 19 peaks.

The transmission coefficient for 20 barriers has several other interesting features as well. To mention just two, we note first that the distribution of peaks in Fig. 4(a) provides a striking visualization of the density of states in the first energy band. As can be seen in the figure, the den-

sity of states is low in the middle, and high at each end, with the highest values at the left end.¹⁰ Second, the envelope traced out by the transmission minima persists unchanged as the number of barriers is increased, and thus represents a feature that will have significance even in an infinite system.

V. SUMMARY

Though Mathematica has been used by others to calculate the energy bands of a one-dimensional crystal with delta function barriers,¹¹ coupling Mathematica with a transfer-matrix approach produces a particularly powerful calculational and educational tool. In particular, the results for multiple barriers suggest interesting possibilities for experimental applications, especially when one considers the relevance of periodic potentials to systems such as semiconductor superlattices and CCD devices. Furthermore, the close analogy between tunneling in quantum systems, and evanescent waves in light undergoing total internal reflection,¹² opens the possibility of exceedingly selective optical filters.

Yet, even in the absence of practical applications or experimental devices, the results given here can help to further a student's understanding of, and interest in, quantum mechanics. And with the aid of Mathematica, students can go quickly beyond simple textbook examples to problems of their own design, and hence begin to participate in the pleasures of independent research.

ACKNOWLEDGMENTS

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APPENDIX A: MATRIX TEMPLATES

(* TEMPLATES FOR THE DISCONTINUITY AND PROPAGATION MATRICES *)

```
(* discontinuity matrix *)
d/: d[v1_, v2_] :=
Block[{k1 = Sqrt[e - v1], k2 = Sqrt[e - v2]},
  da = (1 + k1/k2)/2;
  db = (1 - k1/k2)/2;
  Return[{ {da, db}, {db, da} }]
```

```
(* propagation matrix *)
p/: p[v_, length_] :=
Block[{kl = ro*length*Sqrt[e-v]},
  pa = E^(I*kl);
  pb = E^(-I*kl);
  Return[{ {pa, 0}, {0, pb} }]
```

APPENDIX B: UNIT CELL GENERATION

(* GLOBAL VALUES SUPPORTING THE RECTANGULAR BARRIER MODEL *)

```
DIGITS = 10      (* number of digits of interest *)
PRECISION = 40   (* highest precision possible *)
lastArg = 1       (* argument of last eigenvalue; used to track
                   changes in band number because Mathematica
                   reports arguments as negative numbers in the
                   3rd and 4th quadrants *)

band = 0          (* assume start in first band, band 0 *)
period = 1        (* number of barriers in a unit cell *)
delv = 0          (* difference in potential height of successive barriers *)
plotList = {}     (* list of solutions: e.g. {(kd/π, E/V), ... } *)
```

```
ro = 10*Pi/9      (* dimensionless strength of barrier *)
v1 = 0            (* dimensionless (default) reference potential of gap *)
v2 = 1            (* dimensionless (default) potential of barrier *)
lengthBarrier = 1 (* scaling factor for length of barrier *)
lengthGap = 1     (* scaling factor for length of gap *)
```

(* ROUTINES GENERATING TRANSFER MATRICES FOR RECTANGULAR BARRIER MODEL *)

```
(* makeTx generates the matrix that takes the charge across
one barrier and one gap. Both potentials and lengths
are adjustable. *)
makeTx/: makeTx[v1_, v2_, lengthBarrier_, lengthGap_] :=
Block[{tx, p1, p2, d1, d2, a, b, c},
  d1 = d[v1, v2];
  p1 = p[v2, lengthBarrier];
  d2 = d[v2, v1];
  p2 = p[v1, lengthGap];
  tx = N[p2 . d2 . p1 . d1, PRECISION];
  Return[tx]
]

(* makeUnitCell generates the matrix representing 1 unit cell.
the potentials (v1, v2) and the lengths
(lengthBarrier, lengthGap) are global because of
the relative simplicity of the crystal structure
being modeled. *)
makeUnitCell/: makeUnitCell :=
Block[{tx, t, i},
  t = IdentityMatrix[2];
  For[j = 1, i <= period, i++,
    tx = makeTx[v1, v2, lengthBarrier, lengthGap];
    v2 = 2 + delv - v2;
    t = t . tx
  ];
  Return[t]
]
```

APPENDIX C: ENERGY BAND AND TRANSMISSION COEFFICIENT ROUTINES

(* ENERGY BAND DETECTION *)

```
(* SEARCHBAND finds those energy values in the range START to FINISH with
eigenvalues of magnitude 1. The argument of those eigenvalues
is kd, the phase factor by which the propagating wave packet has
shifted .. (see comments to the routine saveArg, below).
STEPS is the resolution with which the bands are investigated. *)
searchBand/: searchBand[start_, finish_, steps_] :=
Block[{deleov, i, zeroTol, eVals, cVals}, (* INITIALIZE AND SET UP *)
  plotList = {};
  zeroTol = 1*10^(-DIGITS);
  c = start;
  deleov = (finish - start)/steps;
  eVals = Eigenvalues[makeUnitCell]; (* INITIALIZE lastArg *)
  lastArg = Arg[eVals[[1]]];

  For[j = 1, i <= steps + 1, i++, (* SWEEP THROUGH E/V RANGE... *)
    eVals = Eigenvalues[makeUnitCell]; (* LOOKING FOR EIGENVALUES = 1 *)
    eVals = Abs[eVals];
    If[Chop[Abs[eVals[[1]]] - 1, zeroTol] == 0, saveArg[eVals[[1]]];
    c = c + deleov
  ];
  band = 0; (* RESET band TO 0 *)
]
```

```
(* SAVEARG keeps track of band number by counting sign changes in the
arguments. Saves solutions in plotList. NOTE:
kd is stored as kd/π (kdoPi). negArg and lastArg
are bookkeeping conveniences, required by the way that
MATHEMATICA reports the argument of a complex number. *)
saveArg/: saveArg[z_] :=
Block[{arg, negArg, kdoPi},
  arg = N[Arg[z], DIGITS + 1];
  If[arg >= 0, negArg = 0, negArg = 1];
  If[Sign[lastArg] + Sign[arg] == 0, band++];
  lastArg = arg;
  kdoPi = (arg + (negArg + band)*Pi)/(period*Pi);
  plotList = Append[plotList, N[{kdoPi, c}, DIGITS]];
]
```

(* TRANSMISSION COEFFICIENT CALCULATION *)

```
(* TCOEFF calculates the transmission coefficient of a system with
NUMBEROFBARRIERS barriers, for an incident charge with energy
in the range START to FINISH. STEPS is the resolution with
which the system is investigated. In this routine we generate a
matrix for only 1 repeating segment of the system, and
then raise that matrix to the power specified by the number of
barriers to obtain the transfer matrix for the full system.
The multiplicative inverse of the square of the element in row 2,
column 2, of the transfer matrix is the transmission coefficient. *)
```

```
tcoeff/: tcoeff[start_, finish_, steps_, numberOfBarriers_] :=
Block[{deleov, i, t22, t}, (* INITIALIZE AND SET UP *)
  plotList = {};
  c = start;
  v1 = 0; v2 = 1; lengthGap = 1; lengthBarrier = 1;
  deleov = (finish - start)/steps;
```

```

For[i = 1, i <= steps + 1, i++,      (* SWEEP THROUGH E/V RANGE *)
  t = makeUnitCell;                  (* MATRIX FOR REPEATING SEGMENT *)
  t = MatrixPower[t, numberOfBarriers]; (* RAISED TO THE DESIRED POWER *)
  i22 = t[[2,2]];                    (* ELEMENT IN ROW 2, COLUMN 2 *)
  plotList = Append[plotList, N[{c, (Abs[i22])^2}, DIGITS]];
  c = c + delev;
];
}

```

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